



Progressive failure in carbon fiber composite laminates with drilling-induced delamination

information from ultrasound



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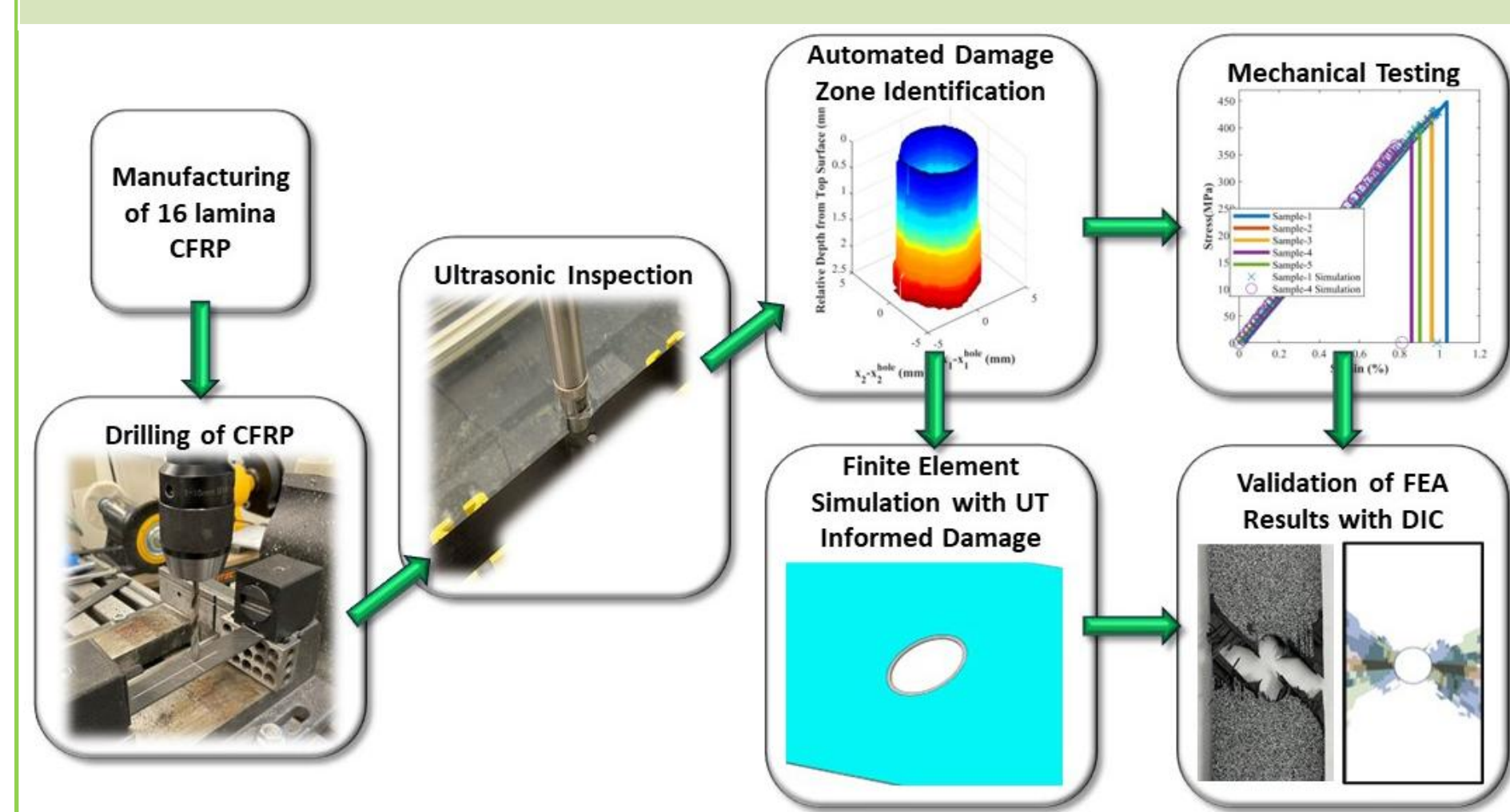
Introduction

Although drilling in composites is often required to facilitate the mechanical joining between parts, it causes delamination, burrs, and chips around the hole rim when worn-off drill bits (improper tool) or high spindle speed and low feed rates (improper setting) are selected. Drilling-induced damage may cause a premature failure of composites, and method to identify accurately the damage to predict the failure is required. Traditionally, a 2-step approach, predicting damage during drilling on a ply-by-ply basis, followed by structural simulation with applied load is employed which is computationally expensive and require validation on both steps.

Objectives: Our study aims to propose a novel framework for:

- Identifying accurately the interlaminar damage (delamination) in composite laminates due to drilling using Ultrasound Scanning system.
- Translating the identified damage into 3D finite element model as delamination in cohesive element,
- Predicting damage progression and failure onset of CFRP laminates using continuum damage mechanics.

Novel framework



Manufacturing, Drilling process and UT Scanning

Panels of UD carbon/epoxy T700/250F of 16-laminae with a quasi-isotropic lay-up $[0^\circ/-45^\circ/+45^\circ/90^\circ]_{2S}$ were manufactured using hot press. Samples of 250mm x 25mm were then drilled using 6.25 mm diameter drill bit (135° tip angle) at 1700 rpm to introduce delamination (Figure-1).



Figure-1. Drilling Process

Identification of delamination using UT

Drilling-induced delamination was scanned using in-house UT immersion system (15MHz transducer) in x_1 (scan), x_2 (index), and x_3 (focus adjustment) directions (Figure-2)

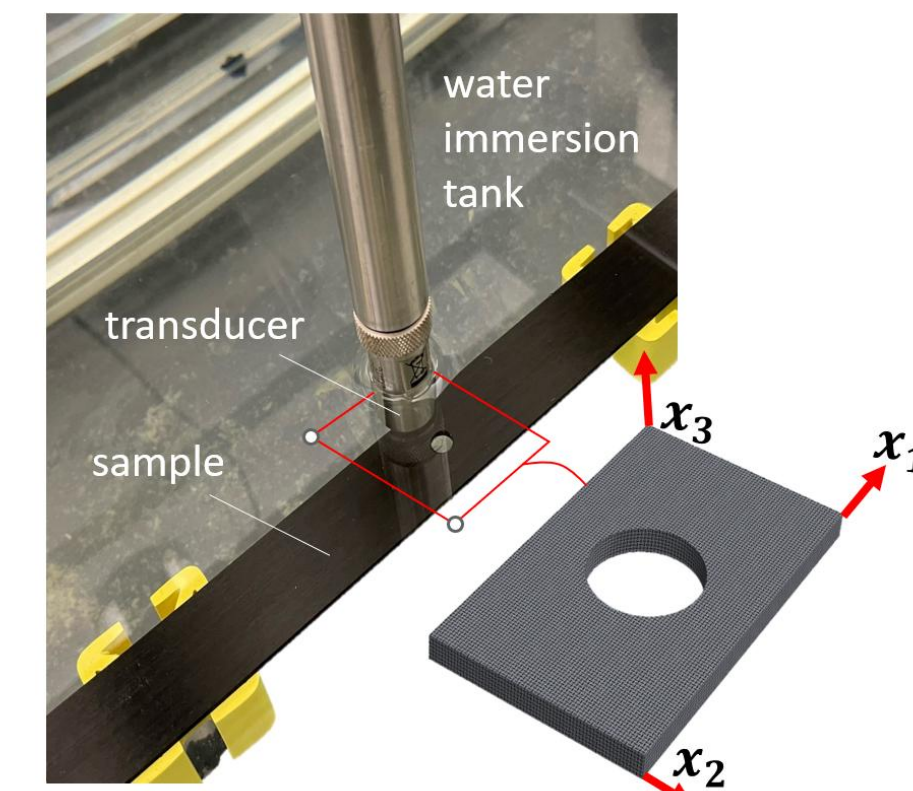


Figure-2. In-house UT immersion System

At each interface, C-scan energy (Figure-3) around the hole are compared to undamaged regions. At same depth, delamination is identified by thresholding waveform's energy relative to undamaged signal resulting in a 3D profile of the damage (Figure-4).

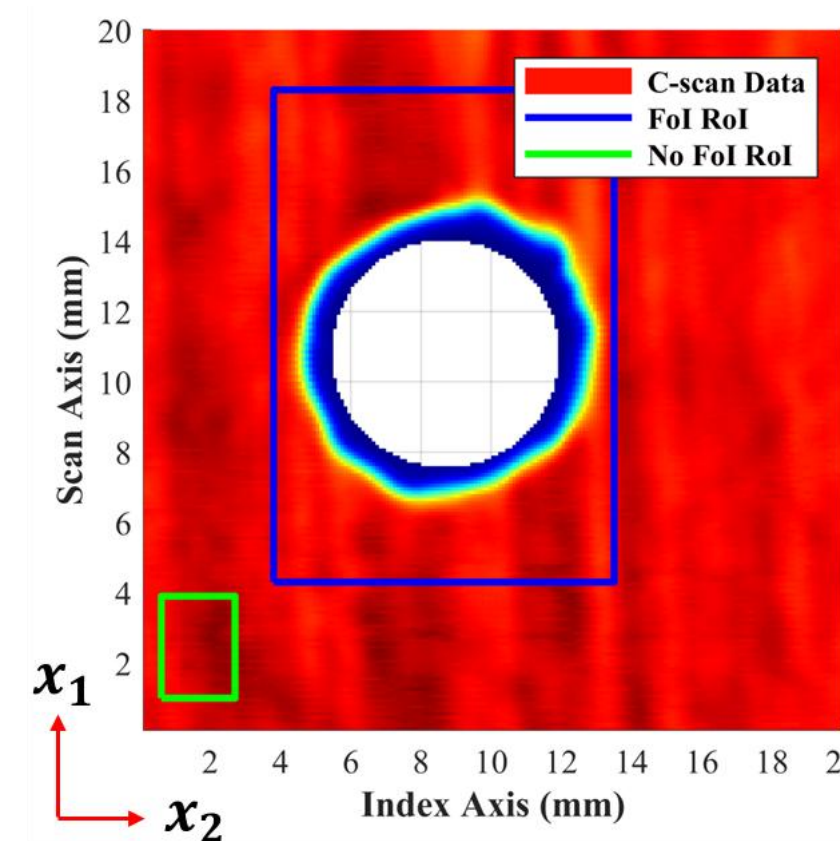


Figure-3. C-scan

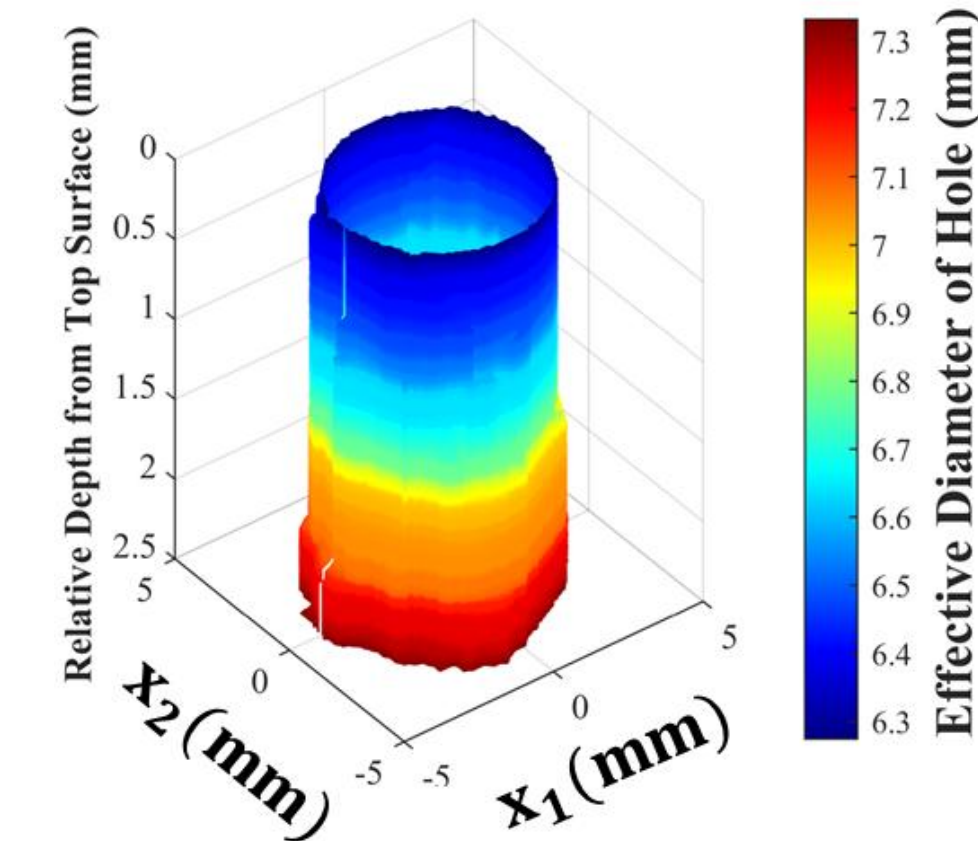


Figure-4. 3D damage area reconstruction

Effective radius (radius of the effective delaminated area) across the thickness of the 5 samples is presented in Figure-5a with damage profile for sample-1 at 3 different depths (b, c, d).

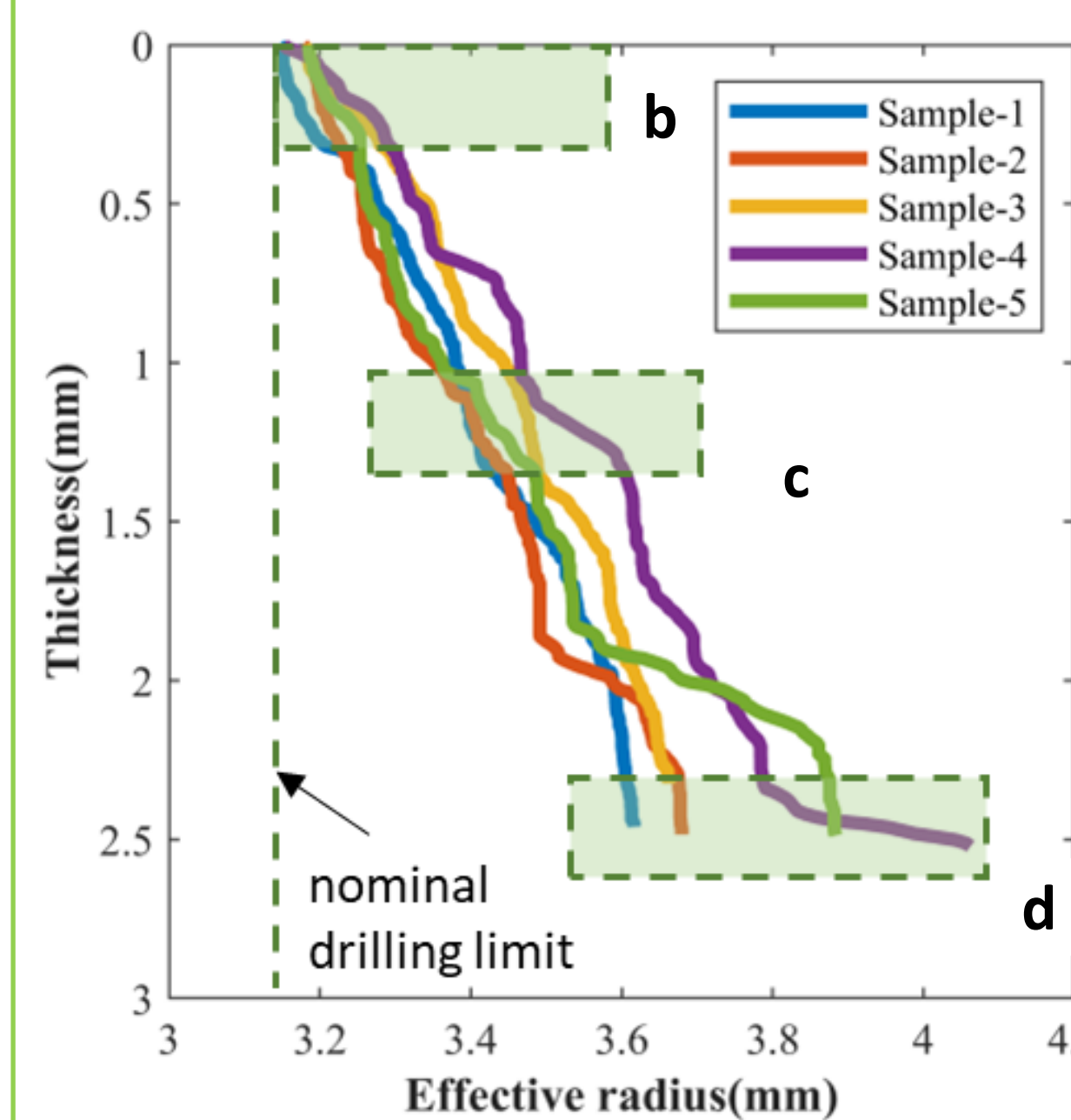
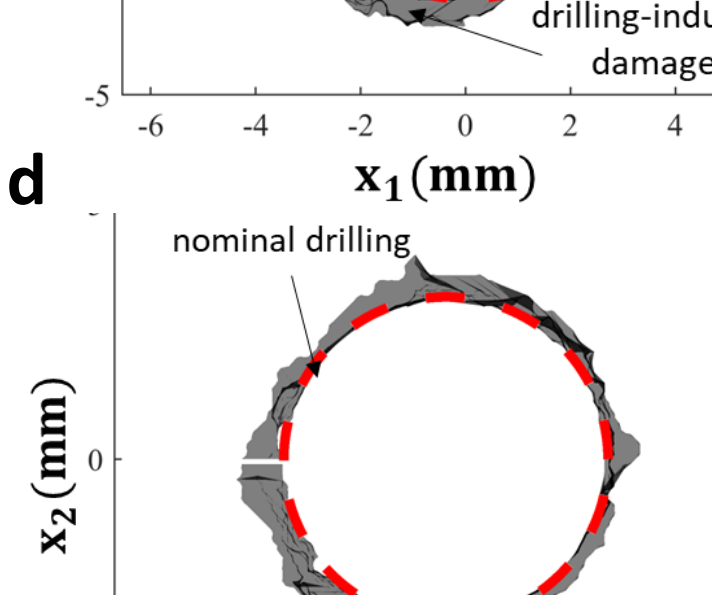
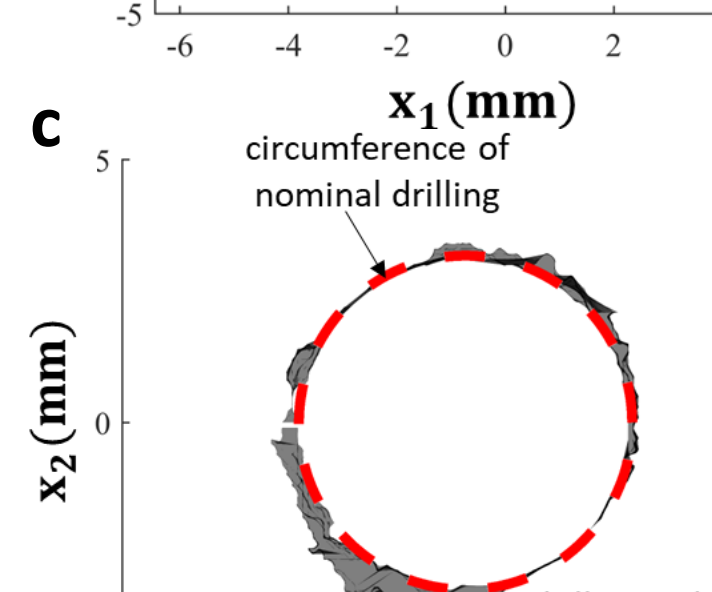
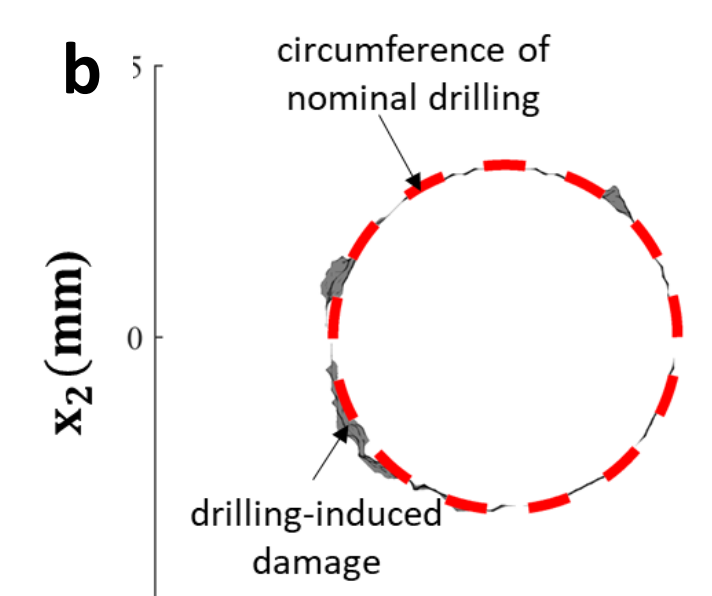


Figure-5. (a) Effective radius variation for 5-samples, Drilling induced damage at 3 different depths (b) 0.1 mm, (c) 1.25 mm, (d) 2.25 mm



Finite Element Method

- Mesoscale model (Figure-6)
- Drilling induced damage geometry information is modeled in cohesive element
- VUMAT was employed for 3D ply Failure criteria with exponential progressive damage evolution law

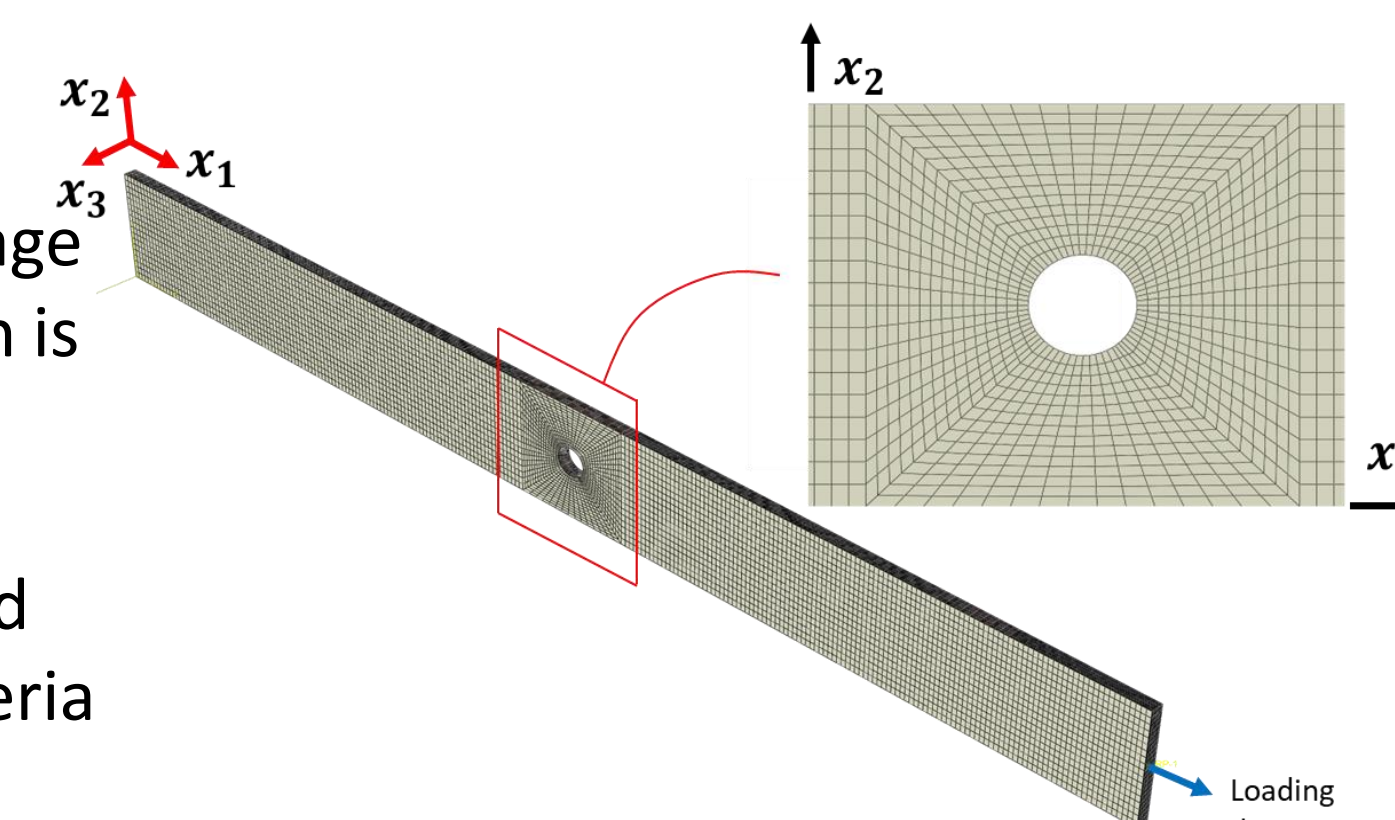


Figure-6. 3D Finite element

Transforming UT data to Finite Element Domain

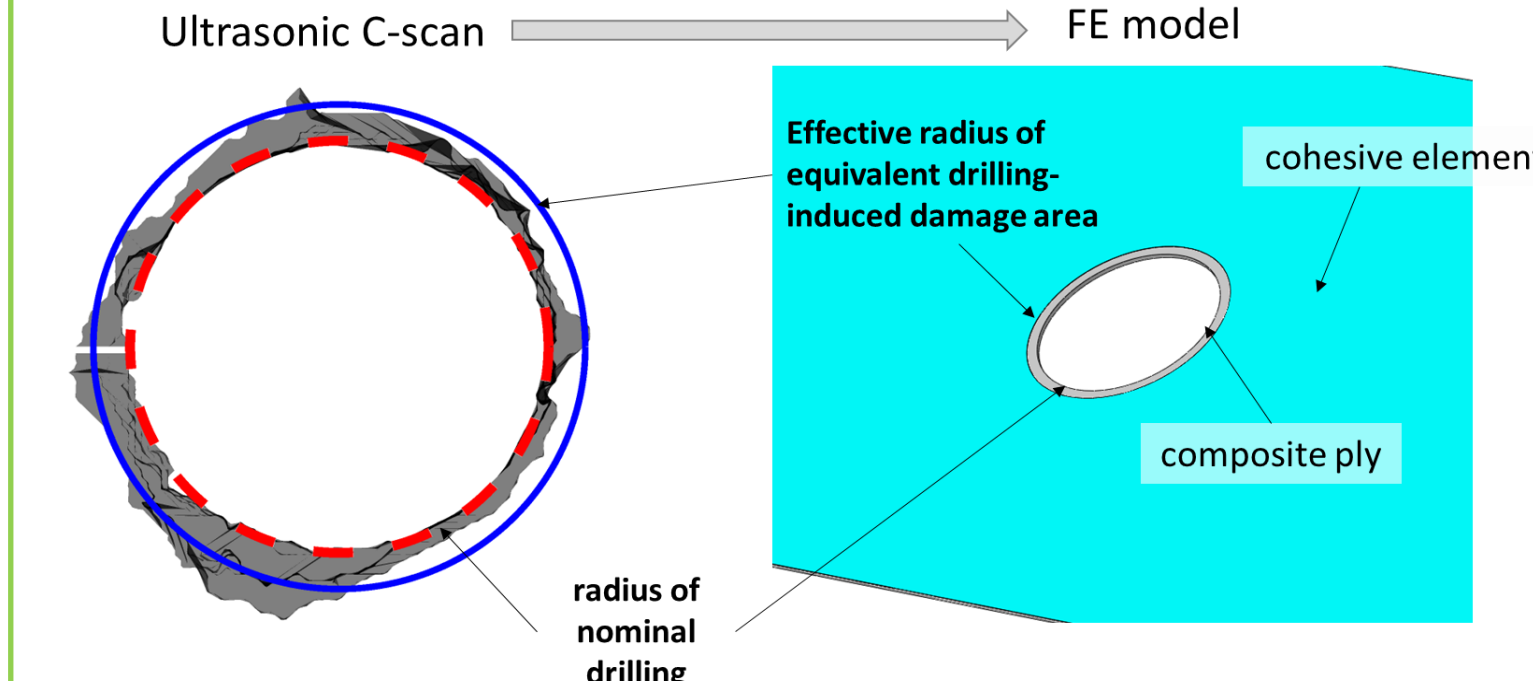


Figure-7. Transforming ultrasonic C-scan data of drilling-induced damage/delamination into FE model

The effective radius of the delaminated interface area in each ply is defined as the effective radius of the hole of cohesive zone (Figure-7)

Mechanical response of open hole tension

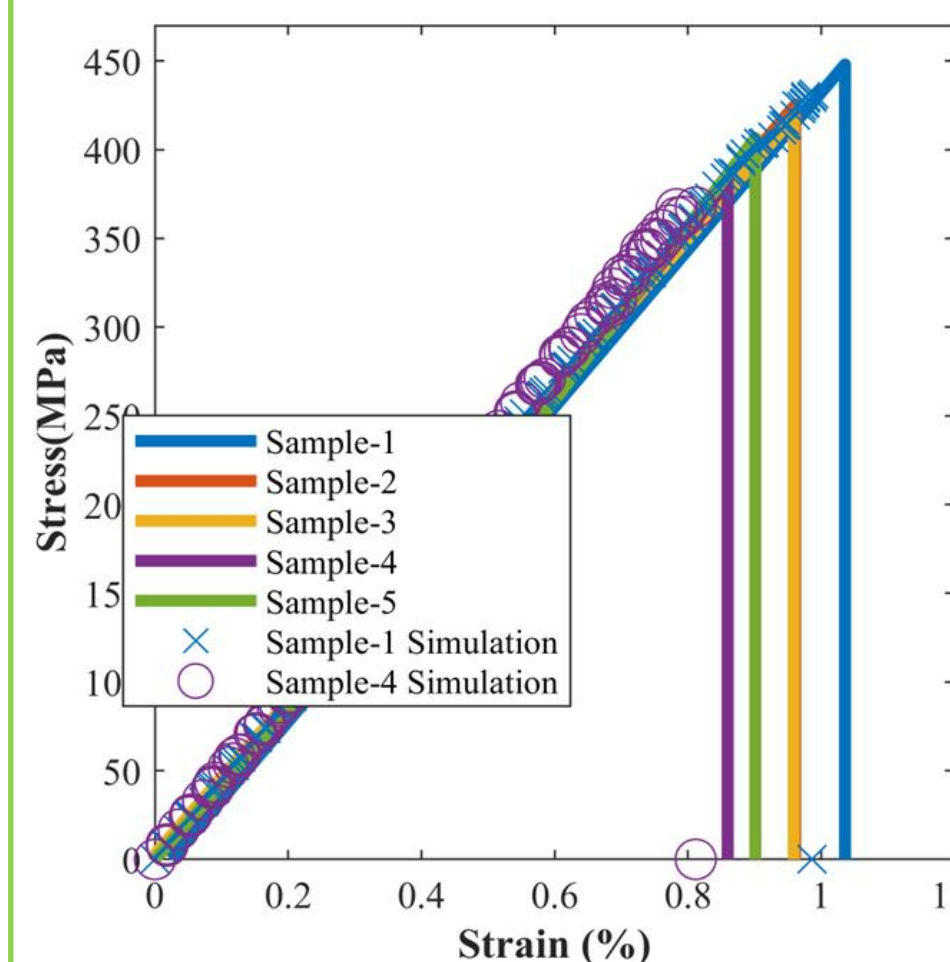


Figure-8. Stress-Strain curves of OHT samples(modelling and experiment)

All five samples exhibit a consistent stress-strain response in open-hole tensile tests (Figure-8), indicating similar effective stiffness at an average of 45.5 ± 2.3 GPa. However, there is notable variability in failure stress, averaging 416 ± 25 MPa, and failure strain at $0.94 \pm 0.07\%$.

Correlation between effective radius and failure strain

Figure-9 indicates that there is a monotonic decrease in the failure strain percentage as a function of increasing delamination area with an average 4.2% error between the experimental and finite element results for the five samples.

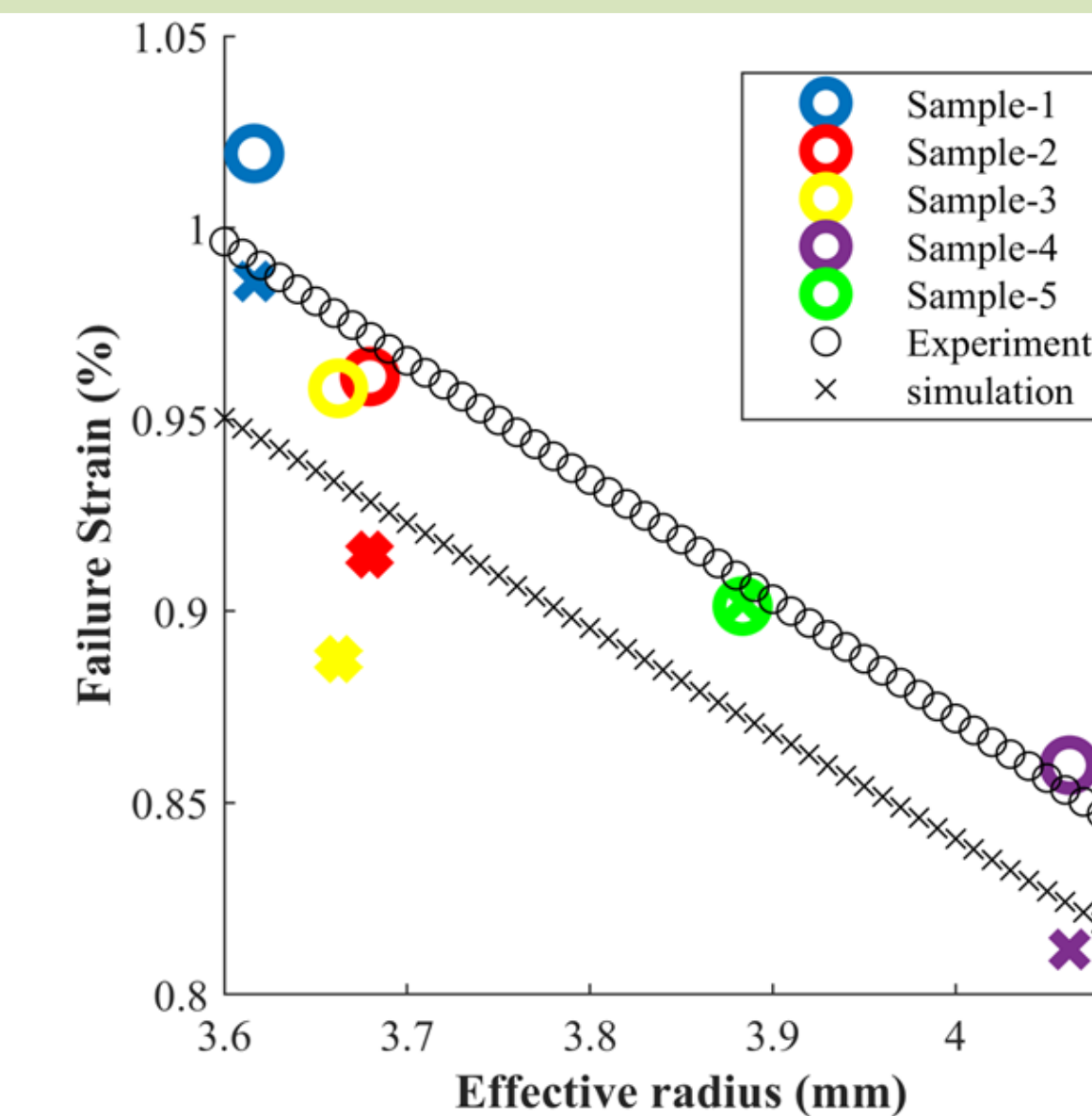


Figure-9. Correlation between failure strain and effective radius

Strain field evolution at the hole rim

The strain field from the experimental observations using digital image correlation is compared to the observed strain field from the finite element simulations for Sample-4 shown in Figure-10.

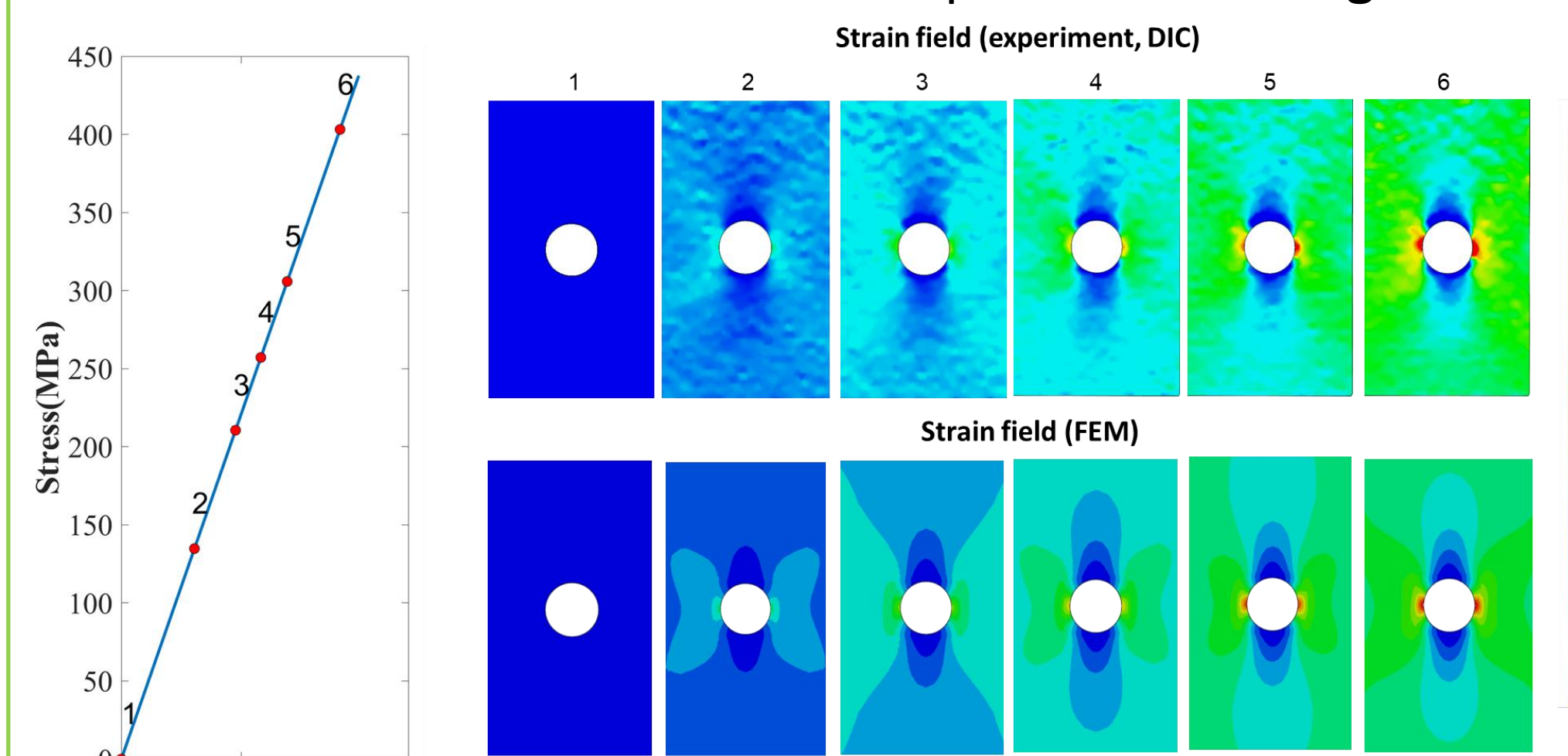


Figure-10. (a) 6 different stress level from Sample-4 OHT test, strain field evolution (b) DIC/Experiment, (c) Finite element method

Stress field evolution at the hole rim

The experimental and simulation stress concentration values are plotted in Figure-11 for sample-5 and the results fall between theoretical (Lekhnitskii) damaged and non-damaged stress distribution indicating the drilling damage does not fully degrade the material rather the damage is only in the interface.

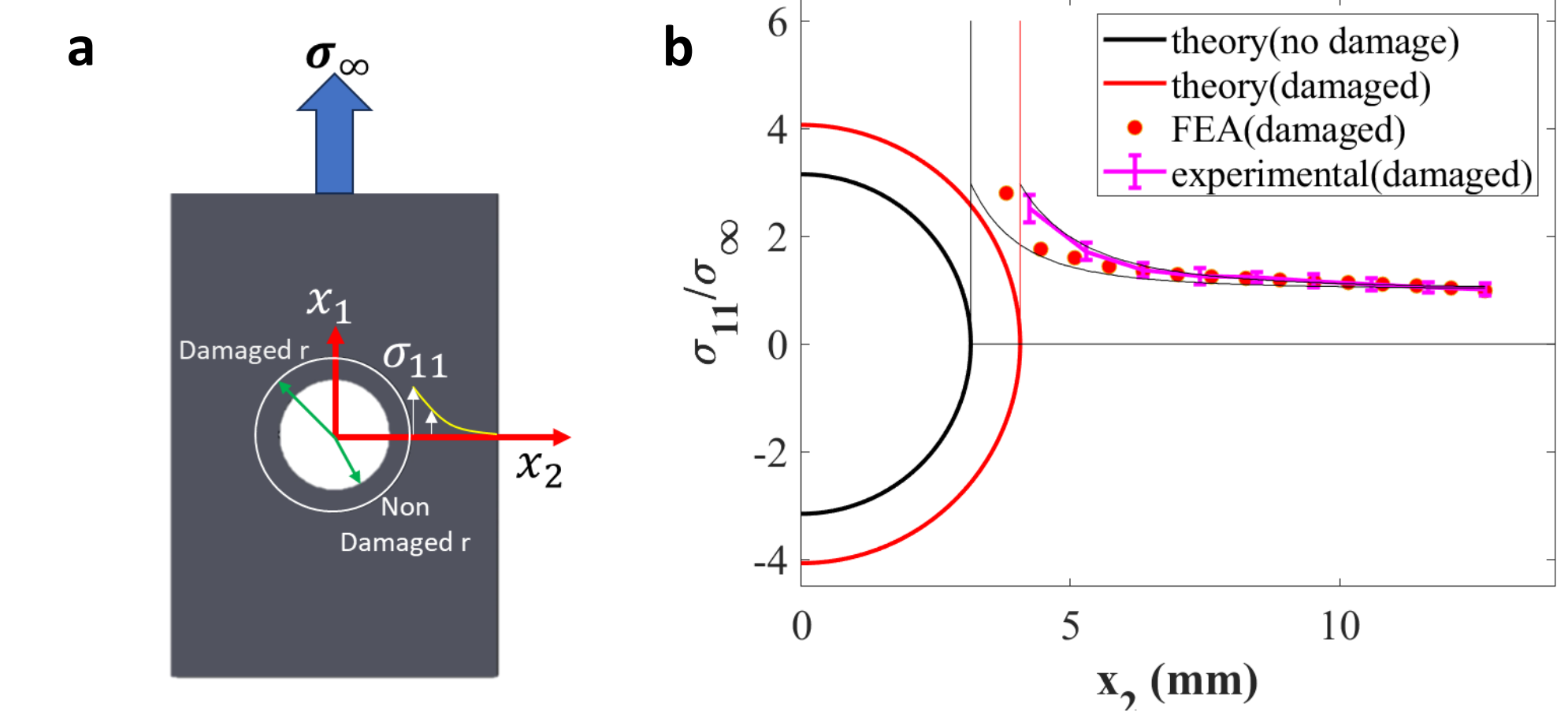


Figure-11. (a) Schematic diagram of OHT test with damaged and non-damaged radii, (b) stress concentration factor from theory, experiment and FEA

Progressive damage

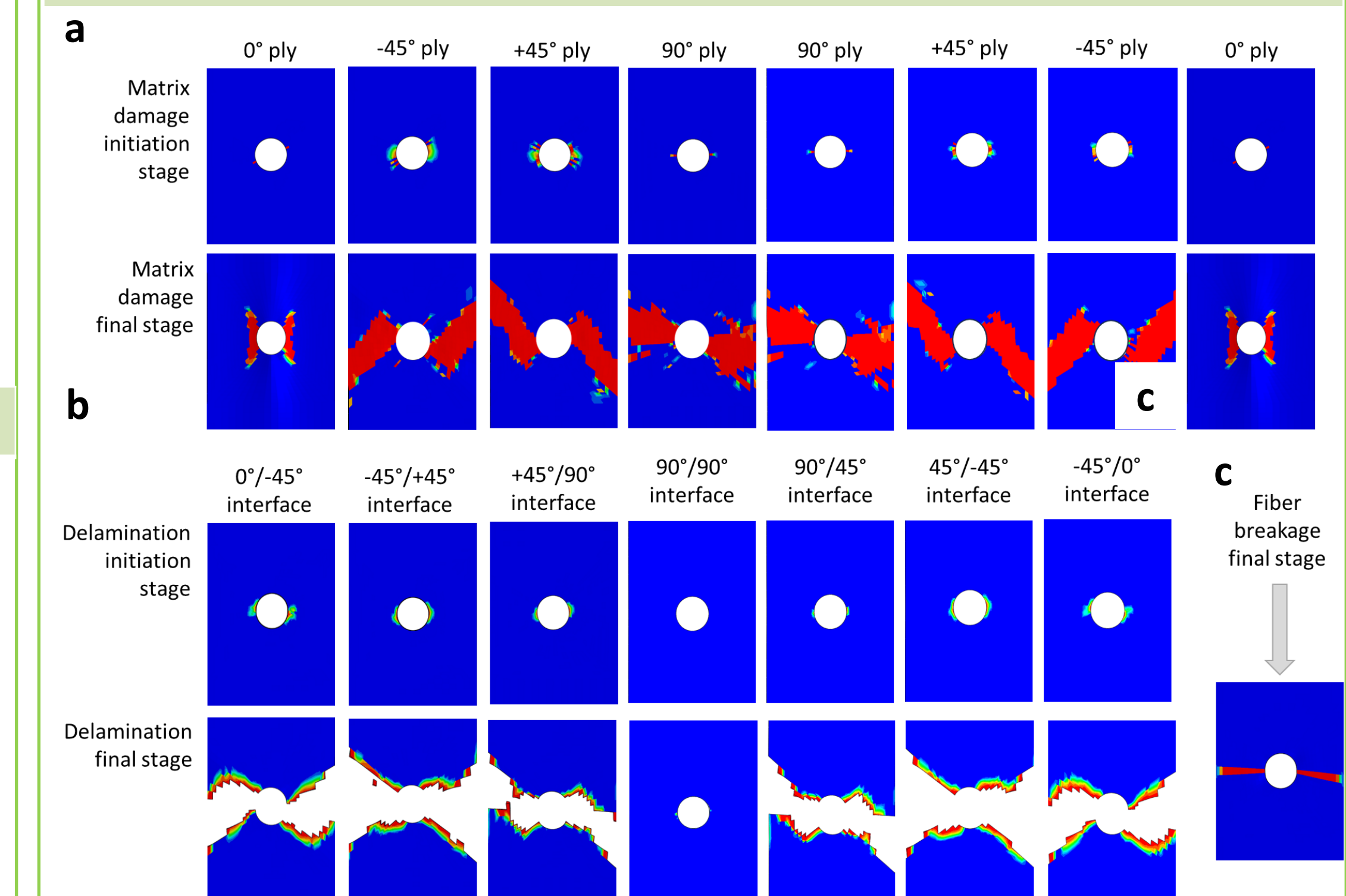


Figure-12. (a) Matrix damage, (b) Interface delamination, (c) Fiber breakage

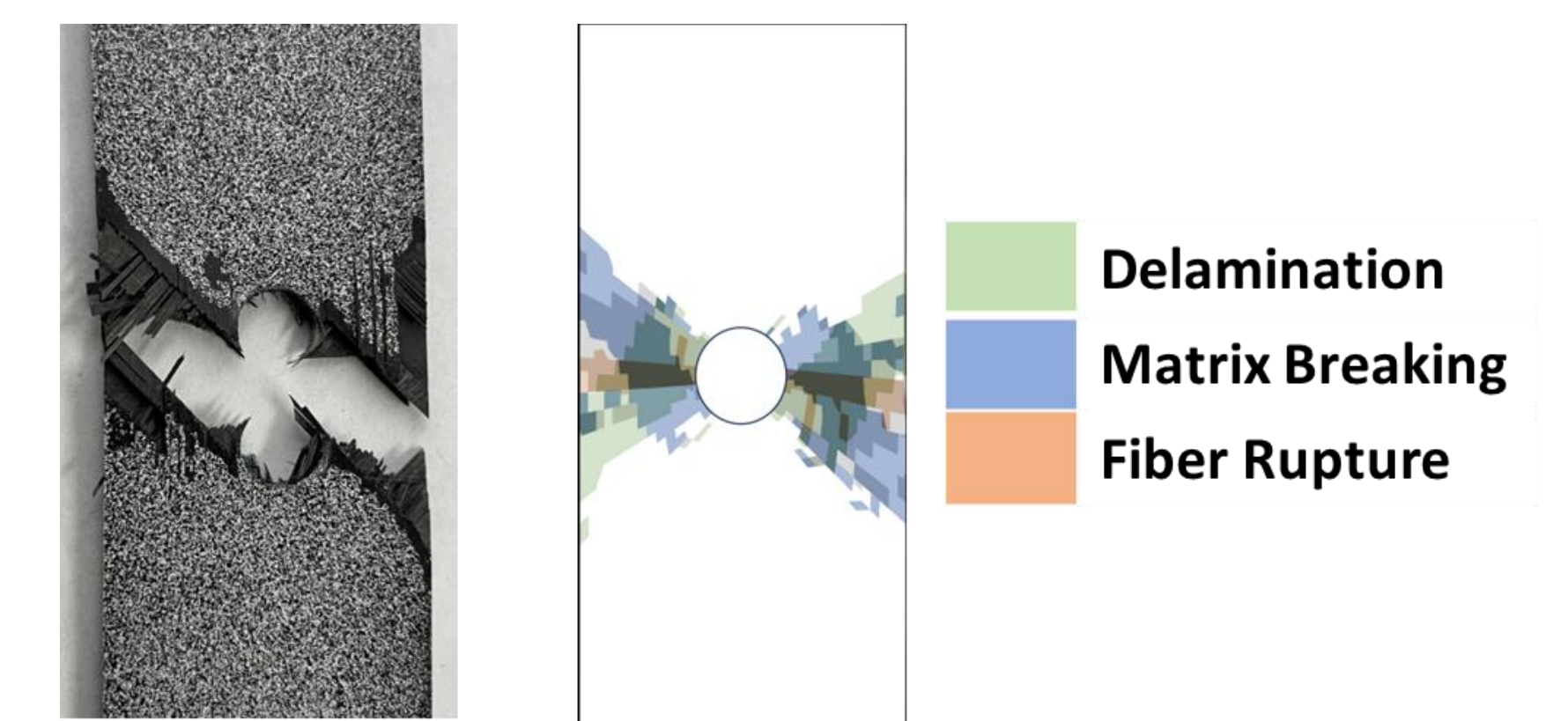


Figure-13. Fracture comparison of open hole model with superposed damage pattern (delamination, matrix breaking, fiber rupture)

Conclusions

We developed a novel framework for identifying interlaminar damage (delamination) in composite laminates due to drilling using UT scan, and for translating the identified damage into 3D finite element model. Cohesive element approach was used to model delamination, while ply degradation was modeled using continuum damage mechanics. The FE model was able to predict the tensile strength and corresponding damage progression during static loading. This new experimental-numerical framework will be useful for simulating similar cases involving UT-identified damage and composite failure.